

Alcohol's Getting a Bit More Social: When Alcohol Marketing Messages on Facebook Increase Young Adults' Intentions to Imbibe

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This study investigates the effects of social media marketing of alcoholic beverages using a 2 (likes: low vs. high) $\times$ 2 (shares: low vs. high) $\times$ 3 (display ad type: alcohol ad vs. antibinge drinking PSA vs. local bank) $\times$ 6 (status update repetitions) experimental design. The study examines how evaluations

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of alcohol marketing status updates and display advertisements predict social media users' intentions to consume alcohol, as a function of message virality and display ad type. Participants' viral behavioral intentions (intentions to like, share, and comment on) for status updates were strongest in predicting intentions to consume alcohol, and this relationship was strongest when the Facebook status update had high likes and shares. The article argues that alcohol marketing on social media reflects a social norm of alcohol consumption, which leads users to consume more alcoholic drinks. Findings are discussed within the framework of persuasion theories and policy changes regarding regulation of alcohol marketing on social media.

INTRODUCTION

Alcohol use and abuse results in 88,000 annual deaths and \$223.5 billion in associated health costs (Centers for Diseases Control and Prevention [CDC], 2014). Young adults, aged 18 to 24, have high risk of alcohol use and abuse (CDC, 2012; National Institute of Alcohol Abuse and Alcoholism [NIAAA], 2013). Not only are young adults the group most likely to consume alcohol, they also are the most frequent and intense binge drinkers. Further, they are highly vulnerable to related advertising and marketing strategies, as demonstrated in past research showing that exposure to alcohol marketing correlates with alcohol consumption (Dunn & Yniguez, 1999; Ellickson, Collins, Hambarsoomians, & McCaffrey, 2005; Garfield, Chung, & Rathouz, 2003; Morgenstern, Isensee, Sargent, & Hanewinkel, 2011; Snyder, Milici, Slater, Sun, & Strizhakova, 2006).

The alcohol industry spent \$3.45 billion on marketing in 2011, with 8% of that amount dedicated to online and digital advertising, recording a fourfold growth from 2005 (Federal Trade Commission, 2014). Although 8% of total marketing budgets may seem minuscule, the affordances of high reach at reduced prices on the Internet (compared to traditional ad spending) signify the magnitude of investment in digital marketing of alcohol. The high risk of alcohol use and abuse among young adults, coupled with their high adoption and use of social media and social networking sites (SNSs; Duggan & Brenner, 2013), elevate the need to investigate the effects of alcohol marketing via social media and SNSs on this vulnerable population.

THE NEW MEDIA MIX

Advertisers see SNSs as increasingly important to the media mix used to reach their audiences, evidenced by 64% of advertisers who planned to

increase their 2013 social media budgets (“Advertisers Boost,” 2013). SNSs are defined as

web-based services that allow individuals to (1) construct a public or semi-public profile within a bounded system, (2) articulate a list of other users with whom they share a connection, and (3) view and traverse their list of connections and those made by others within the system. (boyd & Ellison, 2007, p. 211)

Alcoholic beverage manufacturers have a strong presence on social media. An informal content analysis of the top 25 alcohol brands identified by Siegel et al. (2013) as most consumed by underage drinkers (e.g., Bud Light, Smirnoff Malt Beverages, Budweiser, Coors Light, etc.) showed that more than 90% of these brands have active accounts on Facebook, Twitter, and YouTube. These brands also have a sizable following, with an average of 2.6 million Facebook fans, more than 11,500 followers on Twitter, and more than 2.9 million average video views.

Research on the effects of alcohol marketing has been conducted for decades, but only recently has it begun to explore the phenomena of SNSs. For young consumers—both those just beginning to drink (legally or illegally) and those merely aspiring to do so—social media tend to dominate much of their personal media mix. Moreover, social media are a growing segment of the media mix encountered by more mature consumers.

A decade ago, SNSs were not even known to most consumers, but there has been a dramatic change since then. Edison Research recently found that 56% of all Americans now have a profile on an SNS (Harrison, 2012). Lenhart, Purcell, Smith, and Zickuhr (2010) found that 73% of American teens ages 12 to 17 use SNSs. In addition, as of December 2013, Facebook had more than 1.23 billion active users (Facebook, 2014). SNSs cannot be ignored. Any campaign aimed at reducing alcohol abuse that fails to incorporate these platforms arguably is missing an opportunity to speak to at-risk consumers where they spend much of their media time. Yet little is known about the effects of using SNSs in relation to marketing alcohol.

Empirical research on SNSs related to alcohol is in its infancy. Most studies to date have been descriptive, simply cataloging the features of alcohol marketers’ websites and users’ SNS profiles (Nicholls, 2012; Ridout, Campbell, & Ellis, 2012). One alcohol-related correlational study compared students’ self-portrayals on social media to their self-reported drinking behavior (Glassman, 2012). Still, what is known about the relationship between alcohol consumption and SNS use, particularly causal relationships, remains superficial at best.

eWOM AND VIRALITY

From an advertising perspective, SNSs are generally viewed as electronic forums for “word-of-mouth.” Electronic word of mouth (eWOM), defined as “any positive or negative statement made by potential, actual, or former customers about a product or company, which is made available to a multitude of people and institutions via the Internet,” has potential as a tool to help spread information and excitement about a brand (Hennig-Thurau, Gwinner, Walsh, & Gremler, 2004, p. 39). For advertisers attempting to use this tool, the ultimate objective is to initiate an eWOM chain reaction, where positive recommendations about a brand are rapidly passed from consumer to consumer, known as viral marketing (Liu-Thompkins, 2012). The industry buzzword to measure the effectiveness of such marketing efforts is called “virality.”

Past research defined virality within the scope of access to content (e.g., views and click-through rates; Bonchi, Castillo, & Ienco, 2011; Guerini, Pepe, & Lepri, 2012; Tucker, 2011), eWOM approaches focusing on spreadability and sharing of online messages (De Bruyn & Lilien, 2004; Eckler & Rodgers, 2014; Golan & Zaidner, 2008; Kaikati & Kaikati, 2004; Pastore, 2000; Phelps, Lewis, Mobilio, Perry, & Raman, 2004; Thomas, 2004; Welker, 2002), and through the lens of engagement with online content (Tucker, 2011). To reflect the growing sophistication of the social web, Alhabash and McAlister (2014, p. 3) argued that virality is the combination of users’ online behaviors pertaining to the following:

Viral reach refers to the volume of message viewership, sharing and forwarding by Internet users (e.g., sharing messages on Facebook, re-tweeting on Twitter, etc.).

Affective evaluation refers to users’ explicit emotional responses to online messages. For example, users can “like” a Facebook message, “like” or “dislike” a YouTube video, and make a tweet a “favorite” on Twitter.

Message deliberation deals with Internet users’ active and public deliberation of the message within online social network boundaries (e.g., Facebook comments, Twitter replies). These explicit indicators of deliberation can also reflect the expressed emotional responses to online messages.

Because of its importance, much of the research on virality seeks to unearth what motivates users to interact with messages online (Hennig-Thurau, Gwinner, Walsh, & Gremler, 2004; Strutton, Taylor, & Thompson, 2011). Yet, aside from discovering that relational closeness between sender and receiver (known as tie strength) is a factor, far too little is known about what triggers virality. One possibility is that SNS users discuss or redistribute alcohol messages (pro-alcohol commercials or anti-binge-drinking

PSAs) with which they feel great affinity or that are successful in persuading them to feel positively toward the message.

THE THEORY OF PLANNED BEHAVIOR

Petty and Cacioppo (1986) defined persuasion as the process by which an individual's attitude toward an object, issue, or a person changes. Dual processing models, like the elaboration likelihood model (Petty & Cacioppo, 1986) and the heuristic-systematic model (Chaiken, 1980; Chaiken & Troppe, 1999), argue that an individual's motivation and ability to process information results in either central (systematic) or peripheral (heuristic) route processing. Whereas central or systematic processing requires greater cognitive resources and involves greater elaboration of persuasive arguments, in peripheral or heuristic processing individuals rely on cues and shortcuts to process persuasive messages (Chaiken, 1980; Chaiken & Troppe, 1999; Chen & Chaiken, 1999; Petty & Cacioppo, 1986).

Dual process models explain how individuals undergo attitude formation or change as a result of processing information and assume attitudes are strongly related to behaviors. On the other hand, the theory of planned behavior (TPB; Ajzen & Fishbein, 1980), derived from the theory of reasoned action (Ajzen, 1991), suggests that attitudes are not directly related to behaviors, but rather this relationship is mediated by behavioral intentions (Ajzen, 1991; Ajzen & Fishbein, 1991). TPB posits attitudes, subjective norms, and perceived behavioral control predict behavioral intentions and actual behaviors (Ajzen & Fishbein, 1980). In the current study, we limit our investigation to the relationship between attitudes and social norms, on one side, and behavioral intentions, on the other. We do not investigate the effect of perceived behavioral control (self-efficacy to perform the behavior), previously found to significantly affect behavioral intentions (Ajzen & Fishbein, 1980). We do, however, add another dimension to the TPB by focusing on viral behavioral intentions side by side with attitudes and social norms.

ALCOHOL-RELATED ATTITUDES AND VIRAL BEHAVIORAL INTENTIONS

Attitudes are defined as affective evaluations about individuals (self and others), objects, and issues that vary on two independent dimensions of emotional valence (positive and negativity) and arousal (Maio & Haddock, 2007; Petty & Wegener, 1998; Petty, Wegener, & Fabrigar, 1997). Per TPB, attitudes can be evaluations toward persuasive messages (e.g., advertisement)

or the behavior in question (Ajzen & Fishbein, 1980). Subjective norms refer to perceived social pressures to engage in or avoid particular behaviors and are usually enforced by family, friends, or loved ones (Ajzen & Fishbein, 1980; Cialdini, Reno, & Kallgren, 1990). For example, a young person may have a favorable attitude toward consuming an advertised alcoholic beverage, but due to social pressure to not engage in underage drinking (i.e., parental disapproval), he or she does not actually consume the product. Perceived behavior control, on the other hand, refers to an individual's belief in his or her ability to control performance of a behavior (Ajzen & Fishbein, 1980). An individual may form a negative attitude toward drinking because an antidrinking PSA is effective but nevertheless may engage in drinking because the social norm suggests that drinking is normal, making abstinence difficult to manage.

In the current study, we expose participants to alcohol marketing status updates shared on the Facebook page of an unfamiliar alcohol brand. We are primarily interested in what facets of engaging with an alcohol marketing message posted on a popular SNS lead to the development of intentions to consume alcohol. In addition to the development of attitudes toward the persuasive message (Facebook status update), we also introduce the concept of viral behavioral intentions (VBIs). VBIs are defined as individuals' intentions to engage in behaviors that would make a message viral. These behaviors include liking, sharing, and commenting on messages. Thus, we make a distinction between two types of behavioral intentions: one that is carried out online and does not have direct potential effects on an individual's health (VBI) and another that could potentially affect an individual's health (intentions to consume alcohol). We argue that development of favorable attitudes toward an alcohol marketing Facebook status update, along with developing the intentions to like, share, and comment on it, will significantly increase intentions to consume alcohol. In the context of the current study, a Facebook status update refers to a text message posted by an alcohol company on its own Facebook page.

With regard to attitudes toward the status update, past research on TPB showed that message evaluations positively predicted behavioral intentions (Ajzen & Fishbein, 1980). With limited research linking VBIs to offline behavioral intentions, we posit that expressing intentions to like, share, and comment on the status update signals acceptance of the message and its argument, and thus will positively affect the expression of offline behavioral intentions. Based on this, we hypothesize the following:

- H1: Participants' (a) attitudes toward the status update (A_{SU}) and (b) viral behavioral intentions toward the status updates (VBI_{SU}) will predict intentions to consume alcohol.

We also vary the type of display advertisements accompanying each Facebook status update by including a pro-alcohol advertisement, an anti-binge-drinking PSA, and a control condition featuring a local financial institution. Past research has explored the effectiveness of online display advertising primarily in relation to behavioral intentions and behaviors related to the advertised product, object, or issue (Berthon, Pitt, & Watson, 1996; Burns & Lutz, 2006; Ducoffe, 1996; Goldfarb & Tucker, 2011; Ha, 2008). The current study explores how display advertising can affect outcomes related to the primary messages—the alcohol marketing status updates, in this case. We ask the following questions:

- RQ1: How will participants' (a) attitudes (A_{AD}) and (b) viral behavioral intentions (VBI_{AD}) toward the display ads predict intentions to consume alcohol?
- RQ2: How will participants' (a) A_{SU} , (b) VBI_{SU} , (c) A_{AD} , and (d) VBI_{AD} predict the intentions to consume alcohol, as a function of the type of display ad?

SOCIAL NORMS REGARDING DRINKING

Numerous social psychological theories support the notion that one's attitudes and behaviors are guided by both the behaviors exhibited by others in one's environment and how they are perceived (i.e., condoned or shunned). Social norms are a prominent part of social learning theory (Ajzen, 1991), social cognitive theory (Bandura, 1986), and social comparison theory (Festinger, 1954). Cialdini et al. (1990) identified two types of social norms, descriptive and injunctive. Descriptive norms refer to the perceived prevalence of a particular behavior or attitude in a group or society (e.g., knowing that 80% of college students drink leads to more alcohol consumption), whereas injunctive norms refer to perceived acceptability of behaviors or attitudes (e.g., knowing that 75% of college students think drinking alcohol is fun would prompt students to consume more alcohol).

With four in five college students drinking and half of them binge drinking (NIAAA, 2013), alcohol consumption is considered "the norm" among young adults in America both in terms of descriptive and injunctive norms. This is especially important, because perceptions of alcohol consumption being "normal" have been shown to influence actual alcohol consumption. Borsari and Carey (2000, 2003) demonstrated that college students typically overestimate their peers' alcohol consumption, and this misperception impacts their own drinking behavior.

Other studies have focused on disseminating descriptive norms about drinking to correct overestimation of peers' drinking behaviors (Glazer, Smith, Atkin, & Hamel, 2012; Neighbors et al., 2012; Perkins, Linkenbach, Lewis, & Neighbors, 2010; Thombs & Hamilton, 2002). Lewis and Neighbors (2006) found that when information about norms was effective in correcting misperceptions, this was sufficient to facilitate change in college students' own attitudes toward drinking and actual drinking behavior. Larimer, Turner, Mallett, and Geisner (2004) also found that descriptive norms influence college students' current drinking behavior, whereas injunctive norms are important for influencing planned future drinking behavior.

A number of factors have been found to moderate the effects of social norms on performing behaviors. For example, the amount of attention paid to norms can play an important role in the relationship between norms and actual behaviors, as can the clarity of the normative information, and believability of disseminated information about norms (Cialdini et al., 1990; Park, Smith, Klein, & Martell, 2011). Regarding alcohol consumption, Cullum, O'Grady, Armeli, and Tennen (2012) found that the discrepancy between one's own drinking behavior and the perceptions of others' drinking influenced participants' drinking behavior. Individuals consumed more alcohol when they thought others consumed more, on average, than they did. Of interest, though, the opposite was not true. Individuals did not reduce their alcohol consumption when the average level of consumption of group members was perceived lower than their own.

In the online environment, a message's virality (the number of likes, shares, and comments) could signal social norms surrounding a certain issue or behavior, as it indicates the level of acceptance and prevalence of such issues and behaviors. Virality metrics serve as persuasive cues in affecting people's information selection and processing. As virality signals perceived popularity, it also affects users' decision making and behaviors (e.g., Fu & Sim, 2011; Knobloch-Westerwick, Sharma, Hansen, & Alter, 2005). In addition to measuring virality as a behavioral response—that is, VBI—we argue that message virality can be perceived as a message feature reflecting the social norms related to drinking. In other words, if participants are exposed to a Facebook status update with a higher number of likes and/or shares, they will perceive the message to be normative in reflecting social prevalence and acceptance of the behavior promoted in the status update. As we manipulate the number of likes and shares,¹ we predict that compared

¹In the current study, we limited the manipulation of virality to likes and shares. We did not include comments in our manipulation, as it would entail additional layers of complexity in relation to the features of comments (e.g., valence, length, language, etc.).

to messages with low likes and shares, highly liked and shared alcohol marketing status updates portray greater behavioral acceptance, and will therefore affect participants' desires to consume alcohol. We predict the following:

- H2: The influence of (a) A_{SU} and (b) VBI_{SU} on intentions to consume alcohol will be higher for participants exposed to highly viral status updates (high likes and/or shares) than those exposed to lowly viral status updates (low likes and low shares).

We also explore ways in which attitudes and behavioral intentions toward display advertisements affect intentions to consume alcohol. Due to limited research in this area, we ask the following:

- RQ3: How will the influence of (a) A_{AD} and (b) VBI_{AD} on intentions to consume alcohol vary as a function of virality (the number of likes and shares)?

Finally, we explore how the interaction between display advertisement type and virality affect the relationship between attitudes, VBIs, and intentions to consume alcohol. We ask the following:

- RQ4: How will the influence of (a) A_{SU} , (b) VBI_{SU} , (c) A_{AD} , and (d) VBI_{AD} on intentions to consume alcohol vary as a function of the interaction between display ad type and virality (the number of likes and shares)?

METHOD

Study Design and Participants

The current study employs a 2 (likes: low vs. high) $\times 2$ (shares: low vs. high) $\times 3$ (display ad type: alcohol ad vs. anti-binge-drinking PSA vs. local bank) $\times 6$ (status update repetitions) mixed design, with all factors manipulated between subjects, except repetition. Stimuli comprise various status updates shown with their corresponding likes and shares, alongside display ads. Each status update was written in one of two styles of persuasive appeal: informational and transformational. Although persuasive appeal is not analyzed as a separate independent variable in this study, these different message types increase ecological validity of the stimuli. The present article focuses on the effects of attitudes and viral behavioral intentions on drinking intentions as a function of virality and display ad type.

Participants ($N = 413$) were recruited from introductory classes at a large U.S. midwestern university and were awarded extra credit for participation. Participants were mostly female (57.1%), freshmen and sophomores (63.7%), and White/Caucasian (77.2%), with a mean age of about 21 years ($M = 20.58$, $SD = 1.52$). A conservative criterion was used to retain data from participants whose data points were within 2 standard deviations of the mean on all variables. This resulted in the deletion of 34 outlier cases, some of which had outlying data points on more than one variable (A_{SU} had 16 outliers, VBI_{SU} had four outliers, and A_{ad} had 19 outliers). Hence, the final sample used in analyses comprised 379 cases.

Independent Variables

Display ad type. Display ads were placed alongside alcohol-related Facebook status updates. Each display ad featured a pro-alcohol message (happy hour at a local restaurant), a localized anti-binge-drinking PSA, or a local financial institution. A local bank was used as a control condition because it is highly familiar to the sample and because, unlike a local restaurant, it does not promote or serve alcohol. This factor was manipulated between subjects.

Likes and shares. These two factors were manipulated between subjects. Participants were exposed to status updates in the following conditions: low likes and low shares, low likes and high shares, high likes and low shares, or high likes and high shares.

Status update repetitions. Participants, across all conditions, saw six alcohol-related Facebook status updates presented as screenshots taken from the Facebook page of a rum brand (see Stimuli section). Half of these had an informational appeal, and the other half had a transformational appeal.

Dependent Variables

Attitudes toward the ad (A_{ad}) and status update (A_{SU}). Participants rated each display ad and Facebook status update using three 7-point semantic differential scales (Choi, Miracle, & Biocca, 2001; Coulter & Punj, 2007; MacKenzie & Lutz, 1989; Muehling, 1987). Upon satisfactory factor and reliability analyses (see the appendix), A_{ad} and A_{SU} were computed for each message.

Viral behavioral intentions for ads (VBI_{ad}) and status updates (VBI_{SU}). Participants indicated their agreement/disagreement with

statements using 7-point scales anchored with *strongly disagree* and *strongly agree*. VBI_{ad} and VBI_{SU} shared five items (Alhabash et al., 2013). Two additional items were added to the VBI_{ad} measure. Upon satisfactory factor and reliability analyses (see the appendix), VBI_{ad} and VBI_{SU} were computed for each message.

Intentions to consume alcohol (ICA). Four items were developed to measure participants' intentions to consume alcohol upon exposure to the message and were rated on a 7-point scale anchored by *strongly disagree* and *strongly agree*. Upon satisfactory factor and reliability analyses (see the appendix), ICA was computed for each message.

Stimuli

Stimuli were specifically constructed for this experiment. A panel of advertising and consumer psychology experts comprising professional graduate students and faculty members ($N = 16$) pretested 24 status updates featuring an unfamiliar alcohol brand (Bundaberg Rum—also known as Bundy Rum—which exists in Australia but is unavailable for purchase throughout North America). Twelve messages were predominantly informational and delivered relevant facts to consumers, whereas the other 12 were transformational focused on achieving product acceptance by moving consumers emotionally (Cutler, Thomas, & Rao, 2000). Manipulation of persuasive appeal was achieved through the text of Facebook status updates. Informational messages provided specific product information (e.g., “Bundaberg Red . . . double distilled and full of character. Who would want a rum without character?”), whereas transformational ones focused on winning over viewers with humor or other emotional appeals (e.g., “Bundy makes me feel better, and her look better. Yep, we’re in the business of making lives better ☺”).

In addition, pretest participants rated the positivity, negativity, arousal level, and humor of 18 Facebook display ads² that were constructed for this experiment using InDesign: six featuring a happy hour at local restaurants, six featuring an anti-binge-drinking PSA, and six featuring a local financial institution (control). A high-resolution screenshot of the Bundy Rum Facebook page was used as the basis for creating a mock Facebook page for the brand. Ads were placed to the right side of the screen, and text messages were posted as Facebook status updates. All participants saw the

²Alcohol-related Facebook status updates and display ads were pretested on positivity, negativity, arousal, and humor to ensure that the messages are comparable and that the effects are not contaminated by the emotional tone and appeal in these status updates and ads.

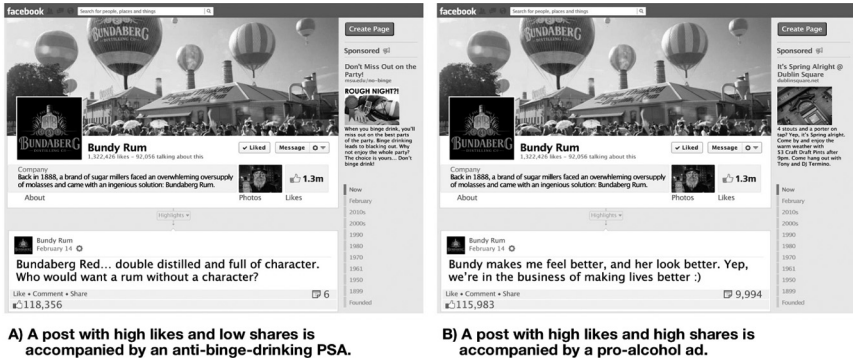


FIGURE 1 Sample experimental stimuli. *Note.* PSA = public service announcement.

same status updates with differences in display ad type of display and virality (see Figure 1).

Procedure

The experimental protocol was approved by the local Institutional Review Board. Participants received a link to an online questionnaire, provided via the Qualtrics online survey service. After providing consent, participants were randomly assigned to conditions. First, they completed unrelated psychological tasks and responded to alcohol and social media use questions. Participants were then exposed to 12 Facebook screenshots in random order. Six of the screenshots were relevant to the manipulations in the current study, and six were distractors (unrelated status updates) that are not included in our analyses. Prior to seeing these screenshots, participants were instructed to take as much time as they like to view each page and then, when they are ready, to click on the “Next” button to answer questions. Participants were not specifically instructed to pay attention to the status updates or the display ads. Following each screenshot, participants answered questions related to dependent variables. A_{SU} questions were asked first, followed by VBI_{SU} , A_{ad} , VBI_{ad} , and ICA. Questions within each of these blocks were presented in a random order.

Finally, participants reported their gender, age, education, ethnicity, and income level. Participants were not asked to respond to questions about whether they noticed the number of likes or shares, or the display ads. This research is focused on the effects of intrinsic message properties on persuasive outcomes, rather than the psychological states elicited by messages. As such, manipulation checks are not required (see O’Keefe, 2003). Participants took, on average, 1 hour to finish the study.

RESULTS

Descriptive Results

Most participants (91.9%) reported that they consume alcoholic beverages at least monthly, of which 17.2% reported consuming alcohol monthly or less, 33.0% drink two to four times a month, 33.4% drink two to three times a week, and 7.3% drink four or more times a week. Most participants reported having a Facebook account (96.3%), where they spend an average of 1.5 hours daily ($SD = 87.40$ minutes) interacting with 749 friends ($SD = 437.76$).

Predicting Intentions to Consume Alcohol

The first set of hypotheses predicted that A_{SU} (H1a) and VBI_{SU} (H1b) would significantly predict intentions to consume alcohol. In addition, RQ1 asked about how A_{ad} and VBI_{ad} would predict intentions to consume alcohol. Data for A_{SU} , VBI_{SU} , A_{ad} , and VBI_{ad} were entered as predictors and the intentions to consume alcohol as criterion in a linear regression model.

The regression model was significant ($R = .70$, $R^2_{adj} = .49$), $F(4, 375) = 89.65$, $p < .001$. In support of H1a and H1b, results showed that A_{SU} ($\beta = .22$, $t = 4.54$, $p < .001$) and VBI_{SU} ($\beta = .51$, $t = 6.64$, $p < .001$) positively predicted intentions to consume alcohol. A_{ad} ($\beta = -.08$, $t = -1.57$, ns) was not a significant predictor of intentions to consume alcohol, whereas VBI_{ad} marginally predicted intentions to consume alcohol ($\beta = .14$, $t = 1.86$, $p = .06$).

Display Ad Type

RQ2 asked about how A_{SU} , A_{ad} , VBI_{SU} , and VBI_{ad} predict intentions to consume alcohol, as a function of Facebook display ad type. Simultaneous regression models for each display ad type condition (alcohol, PSA, control) were run with A_{SU} , VBI_{SU} , A_{ad} , and VBI_{ad} as predictors of intentions to consume alcohol (criterion). As shown in Table 1, all regression models were significant. Results showed that VBI_{SU} was the strongest predictor of intentions to consume alcohol yet with nearly double the standardized beta weight for the PSA and control display ad conditions, compared to the alcohol display ad condition.

Virality Effects

The second set of hypotheses predicted that virality would affect the strength of association between A_{SU} (H2a) and VBI_{SU} (H2b) and the intentions to consume alcohol. In addition, RQ3 asked about how the relationship

TABLE 1
Regression Results for the Effects of Attitudes and VBIs on Intentions to Consume Alcohol,
by Display Ad Type

<i>Variable</i>	<i>Alcohol Ad</i>	<i>PSA</i>	<i>Control</i>
A_{SU}	.24*	.20**	.20*
VBI_{SU}	.30*	.59***	.57***
A_{AD}	-.01	-.11	-.15
VBI_{AD}	.21	.17	.09
Model statistics	$R = .61, R^2_{adj} = .37,$ $F(4, 121) = 17.70***$	$R = .80, R^2_{adj} = .64,$ $F(4, 128) = 57.64***$	$R = .71, R^2_{adj} = .50,$ $F(4, 116) = 28.61***$

Note. VBI=viral behavioral intentions; PSA=public service announcement; SU=status update; AD=attitudes.

* $p < .05$. *** $p < .001$.

between A_{ad} and VBI_{ad} , on one side, and intentions to consume alcohol, on the other, would differ as a function of virality. We ran simultaneous regression models for all virality conditions (low likes/low shares, low likes/high shares, high likes/low shares, and high likes/high shares) with A_{SU} , VBI_{SU} , A_{ad} , and VBI_{ad} as predictors of intentions to consume alcohol (criterion). The regression models were significant for all four conditions (see Table 2).

Results showed that—compared to messages that were not viral (low likes and low shares)—messages that were viral showcased a significant effect of VBI_{SU} on intentions to consume alcohol, which was strongest in all conditions. In addition, A_{SU} was a significant positive predictor in almost all conditions yet strongest in high likes and high shares condition. An interesting case is illustrated in the regression for the low-virality condition, where VBI_{ad} was the strongest predictor of intentions to consume alcohol.

TABLE 2
Regression Results for the Effects of Attitudes and VBIs on Intentions to Consume Alcohol,
by Virality (Likes and Shares)

<i>Variable</i>	<i>Low likes/low shares</i>	<i>Low likes/high shares</i>	<i>High likes/low shares</i>	<i>High likes/high shares</i>
A_{SU}	.25*	.15	.25*	.34***
VBI_{SU}	.21	.45**	.63***	.49***
A_{AD}	-.18	-.11	-.06	-.06
VBI_{AD}	.37*	.31	-.07	.15
Model statistics	$R = .64, R^2_{adj} = .41,$ $F(4, 93) = 16.08***$	$R = .75, R^2_{adj} = .57,$ $F(4, 93) = 30.34***$	$R = .68, R^2_{adj} = .37,$ $F(4, 87) = 18.97***$	$R = .78, R^2_{adj} = .60,$ $F(4, 87) = 23.34***$

Note. VBI=viral behavioral intentions; SU=status update; AD=attitudes. * $p < .05$. ** $p < .01$. *** $p < .001$.

TABLE 3
Regression Results for the Effects of Attitudes and VBIs on Intentions to Consume Alcohol, by Virality (Low vs. High Likes/Shares) & Ad Type
(Alcohol Ad vs. Anti-Binge-Drinking PSA vs. Local Bank Ad)

Variable	Low likes/low shares	Low likes/high shares	High likes/low shares	High likes/high shares
Alcohol ad				
ASU	.47	< .01	.39	.38*
VBI _{SU}	-.24	.07	.55	.69**
A _{AD}	-.57	< .01	.10	.15
VBI _{AD}	.69	.52	-.27	-.16
Model statistics	$R = .56, R^2_{\text{adj}} = .32, F(4, 25) = 2.90^*$	$R = .58, R^2_{\text{adj}} = .34, F(4, 31) = 4.00^{**}$	$R = .65, R^2_{\text{adj}} = .42, F(4, 26) = 4.74^{**}$	$R = .85, R^2_{\text{adj}} = .72, F(4, 24) = 10.09^{***}$
Anti-binge-drinking PSA				
ASU	.31	-.03	.39*	.37**
VBI _{SU}	.16	1.19***	.09	.50*
A _{AD}	-.19	-.08	-.19	-.14
VBI _{AD}	.41	-.24	.54	.11
Model statistics	$R = .72, R^2_{\text{adj}} = .52, F(4, 28) = 5.60^{***}$	$R = .96, R^2_{\text{adj}} = .93, F(4, 28) = 11.90^{***}$	$R = .74, R^2_{\text{adj}} = .55, F(4, 28) = 7.51^{***}$	$R = .85, R^2_{\text{adj}} = .72, F(4, 29) = 8.14^{***}$
Local bank (control)				
ASU	.19	.69**	-.06	.24
VBI _{SU}	.42	.03	.87**	.44
A _{AD}	-.14	-.68*	.04	-.22
VBI _{AD}	.25	.80*	-.18	.21
Model statistics	$R = .67, R^2_{\text{adj}} = .45, F(4, 30) = 6.19^{***}$	$R = .82, R^2_{\text{adj}} = .69, F(4, 24) = 13.05^{**}$	$R = .75, R^2_{\text{adj}} = .56, F(4, 23) = 7.22^{**}$	$R = .71, R^2_{\text{adj}} = .51, F(4, 24) = 6.67^{**}$

Note. VBI = viral behavioral intentions; SU = status update; AD = attitudes; PSA = public service announcement.
* $p < .05$. ** $p < .01$. *** $p < .001$.

Display Ad $\times$ Virality Interaction

RQ4 asked about how A_{SU} , VBI_{SU} , A_{ad} , and VBI_{ad} predict intentions to consume alcohol as a function of the interaction between virality (likes and shares) and display ad type. Similar to previous models, the four predictors were regressed on intentions to consume alcohol for each of the 12 conditions (see Table 3).

Participants exposed to viral alcohol marketing status updates paired with alcohol-related display ads and anti-binge-drinking display ads showed a similar trend in terms of the ways in which A_{SU} , VBI_{SU} , A_{ad} , and VBI_{ad} predict intentions to consume alcohol. Status updates that had low likes and low shares (low on virality) showed no effect of all four predictors on intentions to consume alcohol as a function of display ad type, except for a few marginally significant effects. However, viral status updates interact differently with display ad type.

For status updates paired with alcohol display ads, status updates with high likes and high shares showed a significant effect of A_{SU} and VBI_{SU} on intentions to consume alcohol. Although this is also observed with highly viral messages paired with anti-binge-drinking PSAs, the strongest relationship for VBI_{SU} and intentions to consume alcohol was when messages had low likes and high shares. Generally, results suggest that the relationship between VBI_{SU} and intentions to consume alcohol becomes stronger with greater virality of the status update.

DISCUSSION

Against a backdrop of serious health risks associated with excessive alcohol consumption, this study aimed to determine factors that affect young adults' intentions to consume alcohol after seeing alcohol related messages on Facebook. We find that consumption intentions are higher when participants' attitudes toward alcohol status updates and their viral behavioral intentions toward status updates (e.g., intentions to like, share, and comment on) are more positive. Although the directionality of this relationship is not well understood, it appears that those who intend to consume alcohol and intend to engage in viral behaviors with the alcohol brand's Facebook status update do so, irrespective of subtle messages that might be intended to nudge them in a different direction. It is plausible that the relationship between evaluations and VBIs for pro-alcohol status updates, on one side, and intentions to drink, on the other, is the result of cognitive dissonance reduction strategies participants employ (Festinger, 1962). Developing favorable message evaluations, coupled with intentions to publicly endorse a message through viral behaviors, could be precursors of performing offline behaviors.

Regression results predicting intentions to consume alcohol from attitudes and VBI, as a function of display ad type, show that intentions to consume alcohol is significantly related to VBI_{SU} even in the condition where an anti-binge-drinking message was present. Even more so, A_{SU} and VBI_{SU} 's association with intentions to consume alcohol was strongest when status updates were paired with anti-alcohol PSAs. Possibly, participants in this particular condition experienced psychological reactance, which contributed to a rejection of anti-alcohol messages and acceptance of pro-alcohol ones (Brehm & Brehm, 1981). This is echoed in findings related to the boomerang effects of anti-alcohol, anti-smoking, and anti-drug messages that end up increasing consumption likelihood upon exposure to messages denouncing such behaviors (Hornik, 2006; Ringold, 2002; Wolburg, 2006; Yzer, Capella, Fishbein, Hornik, & Ahern, 2003). Social norms likely play a role here as well. With higher likes and shares, drinking consumption norms are clearly reinforced. Because social norms are so important in driving behavioral intentions, the influence of pro-alcohol status updates outweighs the persuasive appeal of paid display advertisements.

In addition, a similar trend is observed with the control condition (local bank ad). The relationship here, we argue, might be slightly different. The irrelevance of the control condition display ad to the content of the status update may have led participants to focus on status updates more than display ads, especially when it came to evaluating status updates and their intentions to consume alcohol. This finding signals to advertisers the importance of congruence between various messages, that is, alcohol-related advertising paired with high likes and shares resulted in greater VBI than the control condition. An incongruent advertisement from a completely different product or service category appears to be irrelevant, with the status update and its social norms indicators taking precedence. In the context of online social marketing, the findings of this study seems to provide similar implications for advertisers and media planners as they do in traditional venues (e.g., De Pelsmacker, Geuens, & Anckaert, 2002; Zanjani, Diamond, & Chan, 2011): The congruence of ad content and context should also be carefully considered when planning online marketing strategies.

Social norms also appear to play a role in participants' intentions to consume alcohol, consistent with prior research (Glazer et al., 2010; Larimer et al., 2004). In this study, social norms were signaled by high likes and shares, providing cues indicating status updates' popularity; the higher the likes and shares, the greater the indicated strength of the social norm. Virality, as expressed through different combinations of likes and shares displayed underneath the status update, played a significant role in the strength VBI influence on intentions to consume alcohol. Compared to messages that were not viral (i.e., those with low likes and low shares), viral messages had a

significant effect of VBI_{SU} on intentions to consume alcohol. This finding echoes Cullum and colleagues' (2012) findings.

Theoretical and Practical Implications

This study dealt primarily with factors that can predict intentions to consume alcohol, beyond the content of persuasive messages. With the abundance of online communication channels and advertising and marketing strategies, theorizing about persuasion should adapt to this new media environment and further expand the relationship between online and offline behaviors. Our findings show that a considerable amount of variance in the intentions to consume alcohol can be attributed to the likelihood of engaging with content designed for viral distribution online. It is plausible that indicating greater likelihood to engage in viral behaviors (e.g., liking, sharing, and commenting on content shared on SNSs) more strongly primes participants to express intentions to engage in offline behaviors. However, we do realize that the directionality of this relationship requires further investigation, as it is also plausible that those with greater intentions to adopt behaviors offline might be the ones who are more likely to engage in viral behaviors online. In all cases, the new media environment pushes us to acknowledge the dynamic and iterative nature of persuasion. Whereas traditional persuasion models perceived message receivers as passive, our findings provide evidence that the active nature of audience members contributes significantly to how they are affected by persuasive messages. Another theoretically relevant implication provided by our findings deals with the effects of virality on offline behaviors. Our findings confirm the need to redefine social norms within the context of online communication. The volume of affective evaluations (i.e., likes) and viral reach (i.e., shares) subtly influences the online-to-offline behavioral equation.

Taken together, our findings have significant policy implications with regard to regulating alcohol marketing on social media and other socio-technical systems. Whereas regulatory bodies put ample focus on regulating exposure to alcohol marketing messages by underage drinkers, more emphasis should be put in understanding the complex and user-generated nature of social media that extend beyond traditional modes of marketing.

From an advertising perspective, our findings point to the complexity of social media marketing and online advertising in affecting attitudes and behaviors of consumers. It is important to consider this complexity when theorizing about the effects of interacting with persuasive content online. In addition, our findings show that the type of display ads paired with SNS content varies the relationship between attitudes and VBI for status updates and offline behaviors. This also challenges dominant claims about

banner advertising blindness (i.e., highlighted advertisements are most likely to be missed; Benway, 1998), as our findings showed evidence of a possible effect of SNS display advertising on offline behaviors. It is also important to further explore the phenomenon of display advertisement congruence and relevance in line with recent developments in social media advertising.

The effects we found for both attitudes and social norms (virality) on young adults' drinking intentions clearly suggest a challenge for those who may wish to deliver messages promoting responsible drinking. Given that alcohol companies typically have large budgets for creating and disseminating their messages, as well as for hiring paid brand ambassadors (i.e., individuals hired to promote brands with paid-for eWOM), government-sponsored messages promoting responsible drinking are likely to suffer. PSAs, advocacy-group sponsored anti-drinking campaigns, and other messages designed to counteract the prevalence of pro-alcohol messages are in fact the underdog in terms of not having the paid-for power of viral spread.

The absence of some mechanism to show responsible drinking or anti-binge-drinking messages as being normative intensifies perceptions that drinking is "the norm" and alternative views as counternormative. Further, when viewed in concert with Cullum et al.'s (2012) findings, pro-alcohol norms appear to be stronger influencers on behavioral intentions than anti-drinking norms, thus, the challenge becomes even greater to positively influence health behaviors online.

Limitations and Future Research

First, our main dependent variable was a general measure of intentions to consume alcohol, which might hinder the validity of the claim regarding the directionality of the relationship between VBI and drinking intentions. Future research should apply a diverse set of measures to better capture this causal relationship (e.g., intentions to consume the promoted alcohol brands and actual postexposure drinking behavior). Second, we investigated two of TPB's components (attitudes and social norms). Future studies should include a measure of perceived behavioral control to fully test the TPB model. Third, the use of an alcohol brand unknown to our sample to guard against effects of previous brand awareness and loyalty, reliance on self-report measures, and the questioning of participants about their alcohol consumption and social media use prior to stimuli exposure could have influenced the way participants responded to the different measures, and thus should be taken into consideration for future research. Fourth, the sample used in the current study comprised college students from advertising and communication courses. These students were recruited from introductory-level classes and are by no means "experts" in the field of advertising;

however, a sample of individuals outside of the college environment, who also vary in terms of alcohol use and risk for alcoholism, would likely yield more generalizable results. In addition, with a mean participant age of 21 years, the present study includes both legal and underage drinkers. Because recent reports show that underage college students frequently engage in alcohol consumption (e.g., NIAAA, 2013), we wanted to fully represent the population of interest rather than imposing a minimum age requirement for participation. Obviously, the advantage of this approach is that it captures a representative sample of college students across all year levels (freshmen to seniors); however, researchers who are interested solely in behaviors of legal drinkers would do well to replicate this study using a selection criterion of participants being older than 21. Fifth, because the URL for the survey was distributed online, there was an issue with some potential participants visiting the site but not completing the survey. These potential participants were not considered part of the sample in the present study. Researchers who plan to conduct similar studies might consider a lab setting. Having a 1-hour appointment to visit a lab might also alleviate concerns about fatigue if participants know participation time in advance. Finally, there may be other potential factors with significant potential to help us understand online alcohol marketing. For example, the effects of individual differences in motivational activation, sensation seeking (e.g., Lang, Shin, & Lee, 2005), and past drinking behaviors and experiences (e.g., Lee, Maggs, Neighbors, & Patrick, 2011) all introduce variables that need to be addressed for a broader understanding of social media effects on young adults.

Conclusion

With the continued growth of social networking and its burgeoning popularity among youth, understanding how messages on these platforms exert their influence is critical. Social media are increasingly being used to disseminate commercial messages, and the landscape is not as well understood or regulated as traditional media. With respect to risky behaviors such as underage drinking or binge drinking among youth, social media can both exacerbate and help solve the problems. The study described here offers an initial understanding and a foundation for future researchers and policymakers.

REFERENCES

- Advertisers boost social ad budgets. (2013, February 25). eMarketer. Retrieved from <http://www.emarketer.com/Article/Advertisers-Boost-Social-Ad-Budgets-2013/1009688>
- Ajzen, I. (1991). The theory of planned behavior. *Organization Behavior and Human Decision Processes*, 50, 179–211. [http://dx.doi.org/10.1016/0749-5978\(91\)90020-T](http://dx.doi.org/10.1016/0749-5978(91)90020-T)

- Ajzen, I., & Fishbein, M. (1980). *Understanding attitudes and predicting social behaviour*. Englewood Cliffs, NJ: Prentice-Hall.
- Alhabash, S., & McAlister, A. R. (2014). Redefining virality in less broad strokes: Predicting viral behavioral intentions from motivations and uses of Facebook and Twitter. *New Media & Society*. Advance online publication. doi:10.1177/1461444814523726
- Alhabash, S., McAlister, A. R., Hagerstrom, A., Quilliam, E. T., Rifon, N. J., & Richards, J. I. (2013). Between likes and shares: Effects of emotional appeal and virality on the persuasiveness of anticyberbullying messages on Facebook. *Cyberpsychology, Behavior, and Social Networking*, 16, 175–182. doi:10.1089/cyber.2012.0265
- Bandura, A. (1986). *Social foundations of thought and action: A social cognitive theory*. Englewood Cliffs, NJ: Prentice Hall.
- Benway, J. P. (1998). Banner blindness: The irony of attention grabbing on the World Wide Web. *Proceedings of the Human Factors and Ergonomics Society 42nd Annual Meeting*, 42, 463–467. doi:10.1177/154193129804200504
- Berthon, P., Pitt, L. F., & Watson, R. T. (1996). The World Wide Web as an advertising medium. *Journal of Advertising Research*, 36, 43–54. <http://dx.doi.org/10.1017/S0021849996960067>
- Bonchi, F., Castillo, C., & Ienco, D. (2011). Meme ranking to maximize posts virality in micro-blogging platforms. *Journal of Intelligent Information Systems*, 1–29. doi:10.1007/s10844-011-0181-4
- Borsari, B., & Carey, K. B. (2000). Effects of a brief motivational intervention with college student drinkers. *Journal of Consulting and Clinical Psychology*, 68, 728–733. doi:10.1037/0022-006X.68.4.728
- Borsari, B., & Carey, K. B. (2003). Descriptive and injunctive norms in college drinking: A meta-analytic integration. *Journal of Studies on Alcohol*, 64, 331–341.
- boyd, d. m., & Ellison, N. B. (2007). Social network sites: Definition, history, and scholarship. *Journal of Computer-Mediated Communication*, 13, 210–230.
- Brehm, S. S., & Brehm, J. W. (1981). *Psychological reactance: A theory of freedom and control*. New York, NY: Academic Press.
- Burns, K. S., & Lutz, R. J. (2006). The function of format: consumer responses to six on-line advertising formats. *Journal of Advertising*, 35, 53–63. doi:10.2753/JOA0091-3367350104
- Centers for Disease Control and Prevention. (2012, January). *Vital signs: Binge drinking: Nationwide problem, local solutions*. Retrieved from <http://www.cdc.gov/vitalsigns/pdf/2012-01-vitalsigns.pdf>
- Centers for Disease Control and Prevention. (2014, March 14). *Fact sheets—Alcohol use and health*. Retrieved from <http://www.cdc.gov/alcohol/fact-sheets/alcohol-use.htm>
- Chaiken, S. (1980). Heuristic versus systematic information processing and the use of source versus message cues in persuasion. *Journal of Personality and Social Psychology*, 39, 752–766. doi:10.1037/0022-3514.39.5.752
- Chaiken, S., & Troppe, Y. (1999). *Dual-process theories in social psychology*. New York, NY: Guilford.
- Chen, S., & Chaiken, S. (1999). The heuristic-systematic model in its broader context. In S. Chaiken & Y. Troppe (Eds.), *Dual-process theories in social psychology* (pp. 73–96). New York, NY: Guilford.
- Choi, Y. K., Miracle, G. E., & Biocca, F. (2001). The effects of anthropomorphic agents on advertising effectiveness and the mediating role of presence. *Journal of Interactive Advertising*, 2(1). Retrieved from <http://jiad.org/downloadadd8eb.pdf?p=17>
- Cialdini, R. B., Reno, R. R., & Kallgren, C. A. (1990). A focus theory of normative conduct: Recycling the concept of norms to reduce littering in public places. *Journal of Personality and Social Psychology*, 58, 1015–1026. doi:10.1037/0022-3514.58.6.1015

- Coulter, K., & Punj, G. (2007). Understanding the role of idiosyncratic thinking in brand attitude formation. *Journal of Advertising*, 36, 7–20. doi:10.2753/JOA0091-3367360101
- Cullum, J., O'Grady, M., Armeli, S., & Tennen, H. (2012). The role of context-specific norms and group size in alcohol consumption and compliance drinking during natural drinking events. *Basic and Applied Social Psychology*, 34, 304–312. doi:10.1080/01973533.2012.693341
- Cutler, B. D., Thomas, E. G., & Rao, S. R. (2000). Informational/transformational advertising: Differences in usage across media types, product categories, and national cultures. *Journal of International Consumer Marketing*, 12(3), 69–83. doi:10.1300/J046v12n03_05
- De Bruyn, A., & Lilien, G. L. (2004). A multi-stage model of word of mouth through electronic referrals. *eBusiness Research Center*, Working Paper. Retrieved from: http://www.debruy.nl/info/research/papers/working/debruy.nlilien2004_referrals.pdf
- De Pelsmacker, P., Geuens, M., & Anckaert, P. (2002). Media context and advertising effectiveness: The role of context appreciation and context/ad similarity. *Journal of Advertising*, 31, 49–61. doi:10.1080/00913367.2002.10673666
- Ducoffe, R. H. (1996). Advertising value and advertising on the web. *Journal of Advertising Research*, 36(5), 21–36.
- Duggan, M., & Brenner, J. (2013). *The demographics of social media users, 2012*. Washington, DC: Pew Research Center's Internet & American Life Project. Retrieved from http://pewinternet.org/~media/Files/Reports/2013/PIP_SocialMediaUsers.pdf
- Dunn, M. E., & Yniguez, R. M. (1999). Experimental demonstration of the influence of alcohol advertising on the activation of alcohol expectancies in memory among fourth- and fifth-grade children. *Experimental and Clinical Psychopharmacology*, 7, 473–483. doi:10.1037/1064-1297.7.4.473
- Eckler, P., & Rodgers, S. (2014). Viral advertising: A conceptualization. In H. Cheng (Ed.), *The handbook of international advertising research* (pp. 184–202). West Sussex, UK: Wiley & Sons.
- Ellickson, P. L., Collins, R. L., Hambarsoomians, K., & McCaffrey, D. F. (2005). Does alcohol advertising promote adolescent drinking? Results from a longitudinal assessment. *Addiction*, 100, 235–246. doi:10.1111/j.1360-0443.2005.00974.x
- Facebook. (2014). Newsroom: Company Info. Retrieved from <http://newsroom.fb.com/Key-Facts>
- Federal Trade Commission. (2014, March). *Self-regulation in the alcohol industry: Report of the Federal Trade Commission*. Retrieved from <http://www.ftc.gov/system/files/documents/reports/self-regulation-alcohol-industry-report-federal-trade-commission/140320alcoholreport.pdf>
- Festinger, L. (1954). A theory of social comparison processes. *Human Relations*, 7, 117–140. doi:10.1177/001872675400700202
- Festinger, L. (1962). *A theory of cognitive dissonance* (Vol. 2). Stanford, CA: Stanford University Press.
- Fu, W. W., & Sim, C. (2011). Aggregate bandwagon effect on online videos' viewership: Value uncertainty, popularity cues, and heuristics. *Journal of the American Society for Information Science and Technology*, 62, 2382–2395. doi:10.1002/asi.21641
- Garfield, C. F., Chung, P. J., & Rathouz, P. J. (2003). Alcohol advertising in magazines and adolescent readership. *Journal of the American Medical Association*, 289, 2424–2429. doi:10.1001/jama.289.18.2424
- Glassman, T. (2012). Implications for college students posting pictures of themselves drinking alcohol on facebook. *Journal of Alcohol and Drug Education*, 56, 38–58. <http://www.biomedsearch.com/article/Implications-college-students-posting-pictures/298503332.html>
- Glazer, E., Smith, S. W., Atkin, C. K., & Hamel, L. M. (2010). The effects of sensation seeking, misperceptions of peer consumption, and believability of social norms messages on alcohol

- consumption. *Journal of Health Communication*, 15, 825–839. doi:10.1080/10810730.2010.522222
- Golan, G., & Zaidner, L. (2008). Creative strategies in viral advertising: An application of Taylor's six-segment message strategy wheel. *Journal of Computer-Mediated Communication*, 13, 959–972. doi:10.1111/j.1083-6101.2008.00426.x
- Goldfarb, A., & Tucker, C. E. (2011). Privacy regulation and online advertising. *Management Science*, 57, 57–71. <http://dx.doi.org/10.1287/mnsc.1100.1246>
- Guerini, M., Pepe, A., & Lepri, B. (2012). Do linguistic style and readability of scientific abstracts affect their virality. *Proceedings of ICWSM*. Retrieved from <http://arxiv.org/pdf/1203.4238.pdf>
- Ha, L. (2008). Online advertising research in advertising journals: A review. *Journal of Current Issues in Advertising*, 30, 31–48. doi:10.1080/10641734.2008.10505236
- Harrison, J. (2012, August 22). Businesses benefit from the growing popularity of social media. *Interactyx Limited*. Retrieved from <http://interactyx.com/social-learning-blog/businesses-benefit-from-the-growing-popularity-of-social-media/>
- Hennig-Thurau, T., Gwinner, K. P., Walsh, G., & Gremler, D. D. (2004). Electronic word-of-mouth via consumer-opinion platforms: What motivates consumers to articulate themselves on the Internet? *Journal of Interactive Marketing*, 18, 38–52. <http://dx.doi.org/10.1002/dir.10073>
- Hornik, R. (2006). Personal influence and the effects of the national youth anti-drug media campaign. *The ANNALS of the American Academy of Political and Social Science*, 608, 282–300. doi:10.1177/0002716206291972
- Kaikati, J. G., & Kaikati, A. M. (2004). Stealth marketing: How to reach consumers surreptitiously. *California Management Review*, 46(4), 6–22. Retrieved from <http://ssrn.com/abstract=1394975>
- Knobloch-Westerwick, S., Sharma, N., Hansen, D. L., & Alter, S. (2005). Impact of popularity indications on readers' selective exposure to online news. *Journal of Broadcasting & Electronic Media*, 49, 296–313. doi:10.1207/s15506878jobem4903_3
- Lang, A., Shin, M., & Lee, S. (2005). Sensation seeking, motivation, and substance use: A dual system approach. *Media Psychology*, 7, 1–29. doi:10.1207/S1532785XMEP0701_1
- Larimer, M. E., Turner, A. P., Mallett, K. A., & Geisner, I. M. (2004). Predicting drinking behavior and alcohol-related problems among fraternity and sorority members: Examining the role of descriptive and injunctive norms. *Psychology of Addictive Behaviors*, 18, 203–212. doi:10.1037/0893-164X.18.3.203
- Lee, C. M., Maggs, J. L., Neighbors, C., & Patrick, M. E. (2011). Positive and negative alcohol-related consequences: Associations with past drinking. *Journal of Adolescence*, 34, 87–94. <http://dx.doi.org/10.1016/j.adolescence.2010.01.009>
- Lenhart, A., Purcell, K., Smith, A., & Zickuhr, K. (2010). *Social media & mobile internet use among teens and young adults*. Pew Internet & American Life Project. Retrieved from <http://pewinternet.org/Reports/2010/Social-Media-and-Young-Adults.aspx>
- Lewis, M. A., & Neighbors, C. (2006). Social norms approaches using descriptive drinking norms education: A review of the research on personalized normative feedback. *Journal of American College Health*, 54, 213–218. doi:10.3200/JACH.54.4.213-218
- Liu-Thompkins, Y. (2012). Seeding viral content: The role of message and network factors. *Journal of Advertising Research*, 52, 465–478.
- MacKenzie, S. B., & Lutz, R. J. (1989). An empirical examination of the structural antecedents of attitude toward the ad in an advertising pretesting context. *Journal of Marketing*, 53, 48–65. doi:10.1108/13555851311290911
- Maio, G., & Haddock, G. (2007). Attitude change. In A. Kruglanski & E. Higgins (Eds.), *Social psychology: Handbook of basic principles* (2nd ed., pp. 565–586). New York: The Guilford Press.

- Morgenstern, M., Isensee, B., Sargent, J. D., & Hanewinke, R. (2011). Attitudes as mediators of the longitudinal association between alcohol advertising and youth drinking. *Archives of Pediatrics & Adolescent Medicine*, 165, 610–616. doi:10.1001/archpediatrics.2011.12
- Muehling, D. D. (1987). An Investigation of factors underlying attitudes toward advertising-in-general. *Journal of Advertising*, 16, 32–40. doi:10.1080/00913367.1987.10673058
- National Institute of Alcohol Abuse and Alcoholism. (2013). *College drinking*. Retrieved from <http://pubs.niaaa.nih.gov/publications/CollegeFactSheet/CollegeFactSheet.pdf>
- Neighbors, C., Lee, C. M., Atkins, D. C., Lewis, M. A., Kaysen, D., Mittmann, A., . . . Larimer, M. E. (2012). A randomized controlled trial of event-specific prevention strategies for reducing problematic drinking associated with 21st birthday celebrations. *Journal of Consulting and Clinical Psychology*, 80, 850–862. doi:10.1037/a0029480
- Nicholls, J. (2012). Everyday, everywhere: Alcohol marketing and social media-current trends. *Alcohol Alcoholism*, 47, 486–493. doi:10.1093/alcac/ags043
- O’Keefe, D. J. (2003). Message properties, mediating states, and manipulation checks: Claims, evidence, and data analysis in experimental persuasive message effects research. *Communication Theory*, 13, 251–274. doi:10.1111/j.1468-2885.2003.tb00292.x
- Park, H. S., Smith, S. W., Klein, K. A., & Martell, D. (2011). College students’ estimation and accuracy of other students’ drinking and believability of advertisements featured in a social norms campaign. *Journal of Health Communication*, 16, 504–518. doi:10.1080/10810730.2010.546481
- Pastore, M. (2000, June 15). *The value of word of mouth*. Retrieved from <http://www.clickz.com/clickz/news/1703408/the-value-word-mouth>
- Perkins, H., Linkenbach, J. W., Lewis, M. W., & Neighbors, C. (2010). Effectiveness of social norms media marketing in reducing drinking and driving: A statewide campaign. *Addictive Behaviors*, 35, 866–874. <http://dx.doi.org/10.1016/j.addbeh.2010.05.004>
- Petty, E. R., & Cacioppo, E. R. (1986). *Communication and persuasion: Central and peripheral routes to attitudes change*. New York, NY: Springer-Verlag.
- Phelps, J., Lewis, R., Mobilio, L., Perry, D., & Raman, N. (2004). Viral marketing or electronic word-of-mouth advertising: Examining Consumer responses and motivations to pass along email. *Journal of Advertising Research*, 44, 333–348.
- Ridout, B., Campbell, A., & Ellis, L. (2012). “Off your Face(Book)”: Alcohol in online social identity construction and its relation to problem drinking in university students. *Drug and Alcohol Review*, 31, 20–26. doi:10.1111/j.1465-3362.2010.00277.x
- Ringold, D. J. (2002). Boomerang effects in response to public health interventions: some unintended consequences in the alcoholic beverage market. *Journal of Consumer Policy*, 25, 27–63. doi:10.1023/A:1014588126336
- Siegel, M., DeJong, W., Naimi, T. S., et al. Brand-specific consumption of alcohol among underage youth in the United States. *Alcoholism: Clinical and Experimental Research*, 37, 1195–1203.
- Snyder, L. B., Milici, F. F., Slater, M., Sun, H., & Strizhakova, Y. (2006). Effects of alcohol advertising exposure on drinking among youth. *Archives of Pediatrics & Adolescent Medicine*, 160, 18–24. doi:10.1001/archpedi.160.1.18
- Strutton, D., Taylor, D. G., & Thompson, K. (2011). Investigating generational differences in e-WOM behaviours: For advertising purposes, does X=Y? *International Journal of Advertising*, 30, 559–586. Retrieved from <http://www.internationaljournalofadvertising.com/ArticleViewer.aspx?ID=95242>
- Thomas, G. M., Jr. (2004). Building the buzz in the hive mind. *Journal of Consumer Behaviour*, 4, 64–72. doi:10.1002/cb.158
- Thombs, D. L., & Hamilton, M. J. (2002). Effects of a social norm feedback campaign on the drinking norms and behavior of division I student-athletes. *Journal of Drug Education*, 32, 227–244. doi:10.2190/2UYU-6 × 9 M-RJ65-3YYH

- Tucker, C. (2011). *Virality, network effects and advertising*. NET Institute. Retrieved from https://archive.nyu.edu/bitstream/2451/31400/2/11_06.pdf
- Welker, C. B. (2002). The paradigm of viral communication. *Information Services and Uses*, 22, 3–8. Retrieved from <http://www.editlib.org/p/95985>
- Wolburg, J. M. (2006). College students' responses to antismoking messages: Denial, defiance, and other boomerang effects. *Journal of Consumer Affairs*, 40, 294–323. doi:10.1111/j.1745-6606.2006.00059.x
- Yzer, M. C., Cappella, J. N., Fishbein, M., Hornik, R., & Ahern, R. K. (2003). The effectiveness of gateway communications in anti-marijuana campaigns. *Journal of Health Communication*, 8, 129–143. doi:10.1080/10810730305695
- Zanjani, S. H., Diamond, W. D., & Chan, K. (2011). Does ad-context congruity help surfers and information seekers remember ads in cluttered e-magazines? *Journal of Advertising*, 40, 67–84. doi:10.2753/JOA0091-3367400405

APPENDIX

TABLE A1
Factor Loadings, Eigenvalues, Percentage of Variance Explained, and Cronbach's Alpha for Dependent Variables

Variables/items	Factor loadings per screenshot (Status Update/Display Ad)					
	1	2	3	4	5	6
Attitudes toward the status (A _{SU})						
Negative . . . Positive	.729	.595	.649	.684	.578	.689
Bad . . . Good	.875	.927	.901	.912	.871	.903
Unfavorable . . . Favorable	.911	.846	.881	.932	.887	.826
Eigenvalue	2.40	2.24	2.31	2.40	2.20	2.30
% variance explained	80.08%	74.52%	76.87%	80.13%	73.43%	76.51%
Cronbach's α	.875	.827	.849	.875	.818	.846
Attitudes toward the ad (A _{AD})						
Negative . . . Positive	.730	.734	.721	.674	.632	.741
Bad . . . Good	.954	.947	.933	.960	.937	.895
Unfavorable . . . Favorable	.858	.895	.878	.873	.907	.877
Eigenvalue	2.43	2.47	2.42	2.39	2.35	2.40
% variance explained	81.04%	82.31%	80.64%	79.63%	78.42%	80.01%
Cronbach's α	.882	.892	.880	.871	.861	.874
Viral behavioral intentions for status updates (VBI _{SU})						
This SU is worth sharing with others	.888	.930	.926	.920	.938	.940
I would recommend this SU to others	.928	.937	.943	.941	.952	.946
I would like this SU on Facebook	.792	.849	.822	.767	.799	.829
I would share this SU on Facebook	.930	.927	.943	.932	.943	.937
I would comment on this SU	.881	.903	.908	.804	.893	.897

(Continued)

TABLE A1
Continued

<i>Variables/items</i>	<i>Factor loadings per screenshot (Status Update/Display Ad)</i>					
	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>	<i>6</i>
Eigenvalue	4.13	4.31	4.30	4.05	4.28	4.31
% variance explained	82.56%	86.16%	86.06%	81.04%	85.60%	86.27%
Cronbach's α	.945	.959	.956	.937	.956	.959
Viral behavioral intentions for ads (VBI _{AD})						
This ad is worth sharing with others	.883	.906	.914	.832	.917	.903
I would recommend this ad to others	.927	.939	.937	.901	.907	.920
I would like this ad on Facebook	.870	.850	.868	.840	.828	.819
I would share this ad on Facebook	.895	.876	.874	.890	.876	.851
I would comment on this ad	.834	.819	.833	.813	.857	.872
I would visit the ad's website	.902	.868	.880	.881	.877	.856
I would visit the Facebook page of entity advertised in the ad	.886	.858	.891	.878	.880	.852
Eigenvalue	5.72	5.59	5.70	5.46	5.62	5.52
% variance explained	81.70%	79.78%	81.48%	78.03%	80.29	78.83%
Cronbach's α	.962	.957	.961	.952	.958	.954
Intentions to consume alcohol						
Seeing this SC make me want to have a drink	.924	.918	.908	.940	.913	.889
After seeing this SC, alcohol sounds more appealing	.876	.882	.838	.888	.897	.928
Seeing this SC make me want to drink more	.933	.938	.939	.940	.946	.924
Seeing this SC makes me want to get wasted	.806	.839	.836	.824	.854	.793
Eigenvalue	3.35	3.40	3.33	3.42	3.44	3.34
% variance explained	83.69%	84.95%	83.14%	85.50%	86.07%	83.55%
Cronbach's α	.935	.941	.932	.943	.946	.934

Note. SU = status update; AD = attitudes; SC = screen shot.

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